QQ-M-44b
August 24, 1964

SUPERSEDING
Fed. Spec. QQ-M-44a
April 27, 1956

FEDERAL SPECIFICATION

MAGNESIUM ALLOY PLATE AND SHEET (AZ31B)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers magnesium alloy plate and sheet, AZ31B alloy.

1.2 Classification.

1.2.1 *Tempers.* Magnesium alloy plate and sheet shall be furnished in the following tempers, as specified (see 6.2).

Condition O—Fully annealed.
Condition H24—Strain hardened then partially annealed.
Condition H26—Strain hardened then partially annealed.

1.2.2 *Sizes and thicknesses.* Magnesium alloy plate and sheet shall be furnished in the sizes and thicknesses as specified (see 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specifications to the extent specified herein.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 184—Identification Marking of Aluminum, Magnesium, and Titanium.
Fed. Std. No. 245—Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products.
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of his specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Altanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-M-3171—Magnesium Alloy; Process for Corrosion Protection of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

FSC 9535

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MIL-STD-649—Aluminum and Magnesium Products Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the material furnished under this specification shall conform to the requirements shown in table I.

TABLE I—*Chemical composition*

Element	AZ31B
	Percent
Aluminum	2.5 to 3.5
Zinc	0.7 to 1.3
Manganese	0.20 minimum
Silicon	0.10 maximum
Calcium	0.04 maximum
Copper	0.05 maximum
Nickel	0.005 maximum
Iron	0.005 maximum
Total other elements	0.30 maximum
Magnesium	Remainder

Note: Analysis shall regularly be made only for the elements specifically mentioned in table I. If however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total thereof does not exceed 0.30 percent.

3.1.1 The contractor shall furnish an analysis of each melt showing the percentage of each of the elements designated in table I.

3.1.2 Chemical analysis by the contractor of the individual melts may be waived at the discretion of the inspector provided that the producer's method of composition control is acceptable to him or that all material in the lot can be identified as being from melts previously analyzed and found to be in conformance with the chemical composition requirements.

3.2 Mechanical properties. The mechanical properties of the sheet, measured in the longitudinal direction, the direction of rolling, shall be as specified in table II.

3.2.1 The minimum mechanical properties of sheet and plate in the as-rolled temper or for thicknesses not shown in table II or for properties not measured in the longitudinal direction shall be as agreed upon between the producer and procuring agency (see 6.2).

3.3 Tolerances. Plates and sheets shall not vary from the specified dimensions or from true straightness of edge by an amount greater than the tolerances specified for flatness, squareness, thickness, width, length, and lateral bow in Fed. Std. No. 245 for plate, flat sheet, or coiled sheet, as applicable.

3.3.1 Tolerances for sizes of sheet and plate not covered by Fed. Std. No. 245 shall be as specified in the contract or order.

3.4 Identification marking. Marking shall be in accordance with Fed. Std. No. 184. In addition to the commercial alloy number, the number of this specification shall be applied to each sheet and plate.

3.5 Protective treatment. Magnesium alloy plates and sheets shall be either oiled or chrome pickled, as specified, as a final process before shipment. Unless otherwise specified for Army, Navy, and Air Force purchases, the plate or sheet shall be given a chrome pickle treatment in accordance with type I of MIL-M-3171.

3.6 Workmanship.

3.6.1 The material shall be uniform in quality and condition, clean, sound, smooth, commercially flat, and free from severe buckles and other injurious defects. Discolorations resulting from annealing operations, and superficial surface abrasions which are not of sufficient depth to affect adversely the static or dynamic mechanical properties or formability of the material, shall not be cause for rejection.

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TABLE II—*Mechanical properties*

Alloy and temper	Thickness, inch	Tensile strength p.s.i. minimum	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 inches, percent, minimum
			p.s.i. minimum	Extension under load (inch in 2 inches)	
AZ31B-O	0.016-0.500	[1]32,000	.....	.....	12
	0.501-2.000	[1]32,000	.....	.....	10
	2.001-3.000	[1]32,000	.....	.....	9
AZ31B-H24	0.016-0.249	39,000	29,000	0.0129	6
	0.250-0.374	38,000	26,000	0.0120	8
	0.375-0.500	37,000	24,000	0.0114	8
	0.501-1.000	36,000	22,000	0.0108	8
	1.001-2.000	34,000	20,000	0.0102	8
	2.001-3.000	34,000	18,000	0.0095	8
AZ31B-H26	0.250-0.374	39,000	27,000	0.0123	6
	0.375-0.500	38,000	26,000	0.0120	6
	0.501-0.750	37,000	25,000	0.0117	6
	0.751-1.000	37,000	23,000	0.0111	6
	1.001-1.500	35,000	22,000	0.0108	6
	1.501-2.000	35,000	21,000	0.0105	6

[1] Tensile strength shall be 40,000 p.s.i. maximum.

3.6.2 The metal shall be given such uniform treatment as the manufacturer desires to produce material that will conform to the requirements specified, applied in a manner that will produce utmost uniformity. Annealing operations shall be performed on the whole of a product, never on a part only.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Lot.* A lot shall consist of plate or sheet of the same condition and thickness subjected to inspection at the same time.

4.2.2 *Chemical samples.*

4.2.2.1 *Chemical analysis samples.* A sample for chemical analysis shall be obtained from each 4,000 pounds or less of a melt of the alloy when chemical analysis is made at the time of melting. When chemical analysis is made on the finished plate or sheet, a sample for chemical analysis shall be taken from each 1,000 pounds or less of material comprising the lot. Not less than 2 ounces of the sample chips which shall be fine, clean, and free from oil, dirt, grit, and foreign matter shall be analyzed. Samples for chemical analysis shall be prepared in accordance with method 111 of Fed. Test Method Std. No. 151 and shall represent the full cross section of the material. Upon approval the manufacturer's method of composition control may be used in lieu of the methods outlined above.

4.2.3 *Tension test samples.* At least one tension test sample shall be selected to represent each 1,000 pounds or less of sheet or 2,000 pounds or less of plate comprising a lot, except that in no case shall less than two tension tests represent a lot. The tension test samples that represent a given lot shall not be selected from the same plate or sheet.

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4.3 Preparation of specimens.

4.3.1 *Tension test specimens.* Tension test specimens shall conform to the requirements of method 211 of Fed. Test Method Std. No. 151. For material less than 1/2 inch in thickness, a test specimen of type F2 shall be used. For material 1/2 inch or more in thickness, a test specimen of type R1, R2, R3, and R5 shall be used.

4.4 Examination. Each plate and sheet shall be carefully examined to determine conformance with the requirements of this specification with respect to workmanship, marking, and dimensional requirements. Sufficient spot checks shall be made to insure conformance with the tolerances specified. On approval of the procuring agency a system of statistical quality control may be used with sampling in accordance with level II and acceptable quality level (AQL) of 1/5 percent defective in MIL-STD-105.

4.5 Test methods.

4.5.1 *Chemical analysis.* Chemical analysis shall be by standard wet chemical or spectrographic methods in accordance with method 112 of Fed. Test Method Std. No. 151. In case of dispute chemical analysis by standard wet chemical methods shall be the basis for acceptance (see note, table I).

4.5.2 *Tension test.* The tension test shall be conducted in accordance with method 211 of Fed. Test Method Std. No. 151.

4.5.3 *Yield strength.* Yield shall be determined by the extension underload or offset methods in accordance with method 211 of Fed. Test Method Std. No. 151. In case of dispute, the offset method shall be the basis for acceptance.

4.6 Rejection and retests.

4.6.1 *Rejection.* If a test specimen fails to meet any of the tests required by this specification, the lot represented by the samples shall be rejected.

4.6.2 *Retests.* Retests will be permitted in accordance with the requirements of Fed. Test Method Std. No. 151.

4.7 Reheat treatment. Where failure of any lot of material to meet the requirements of this specification is due to inadequate annealing, the material may be reannealed, and resubmitted for test. Only two such reannealings will be allowed.

4.8 Preservation, packaging, packing, and marking. Preparation for delivery shall be examined for compliance with section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 General. Unless otherwise specified, all material shall be separated by condition, size, and thickness and packed in crates or boxes of the size commonly used for shipment. Individual plates or sheets which have been given a chrome pickle surface treatment shall be separated by a noncorrosive, nonhygroscopic paper to protect the finish. Oiled plates or sheets shall not be required to be separated by paper.

5.2 Preservation, packaging, and packing. Preservation, packaging, and packing of the material shall be in accordance with MIL-STD-649 as specified for the applicable level.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* In addition to any special marking specified in the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The material covered by this specification possesses a combination of

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good mechanical properties and forming characteristics at about 650°F. AZ31B is a weldable grade of magnesium sheet with calcium controlled to 0.04 percent maximum. The H24 temper combines good flatness and ductility with high strength. The H26 temper provides a material of maximum hardness. It has, in general, higher strength and less ductility in thickness above 0.250 inch, and flatness may not be quite as good as the H24 temper.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Temper (1.2).
(*c*) Size and thickness (1.3).
(*d*) Minimum mechanical properties for sheet and plate in the as rolled (F) temper (3.2).
(*e*) Mechanical properties and tolerances for material outside the specification limits (3.2 and 3.3).
(*f*) Purpose for which sheet or plate is intended.
(*g*) Whether chrome pickled or oiled plate or sheet is desired (3.5).
(*h*) Levels of packaging and packing required (5.1).
(*i*) Whether a system of statistical quality control is required (4.4).

6.3 The material described herein supersedes the conditions and compositions of QQ-M-44a, dated April 27, 1956 as follows: AZ31A superseded by AZ31B with AZ31B assuming AZ31A properties, as specified in table II.

6.4 Transportation description. The transportation description and minimum weights applicable to this commodity are:

Rail:

Plates or sheets, magnesium metal alloy.
Carload minimum weight 30,000 pounds.

Motor:

Plates or sheets, magnesium metal alloy.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MR

Navy Weps

Air Force—ASD(11)

Reviewer Activities:

Army—MR, MI, MU, EL

Navy—Weps

Air Force—ASD(11) MOA (69)

User:

Army—GL

Preparing activity:

Navy—Weps

Copies of this specification may be purchased for 5 cents each.

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